SAT Math Mastery · Session 5

Equivalent Expressions and Exponential Functions

Time 3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 50 minutes), and a short error log.

Topics equivalent expressions: exponent rules, rational expressions, factoring and identities · radical equations · systems with exponents and radicals · radical functions and domain · exponential growth and decay models · interpreting exponential models · recursive and geometric sequences · successive percent changes

Name / Date _____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. HARD · DESMOS

For all real values of x,

$$\frac{6^{2x+3}}{2^{2x-1}} = k \cdot 9^x$$

where k is a constant. What is the value of k?

2. HARD · DESMOS

$$(x+p)(x+q) = x^2 + bx + c$$

In the equation above, p and q are real constants and b and c are integers. Which of the following must be true?
I. $p+q$ is an integer
II. p^2+q^2 is an integer
III. $\frac{p^2+q^2}{2}$ is an integer

(A) I only

(B) I and II only

(C) II and III only

(D) I, II, and III

3. HARD · DESMOS

For all $x > 1$,

$$\frac{a}{x-1} - \frac{bx+c}{x^2+x+1} = \frac{3x^2+8x+4}{x^3-1}$$

where a, b, and c are constants. What is the value of c?

4. HARD · DESMOS

For $y \neq 0$ and $y \neq \frac{1}{3}$, which of the following is equivalent to

$$\frac{\frac{1}{y^3} - 27}{\frac{1}{y} - 3} \; ?$$

(A) $\frac{1}{y^2} - \frac{3}{y} + 9$

(B) $\frac{1}{y^2} + 9$

(C) $\frac{1}{y^2} + \frac{6}{y} + 9$

(D) $\frac{1}{y^2} + \frac{3}{y} + 9$

5. HARD · DESMOS

The polynomial $p(x) = x^3 - 9x^2 + kx - 12$ has a positive integer constant k. For some positive integer n, $x - n$ is a factor of $p(x)$. What is the sum of all possible values of k?

6. HARD · DESMOS

$$P = \frac{a}{\sqrt[5]{w^2}}$$

The formula above relates the positive quantities P and w, where a is a positive constant. Which of the following expresses w in terms of P and a?

(A) $w = \left(\frac{a}{P}\right)^{5/2}$

(B) $w = \left(\frac{P}{a}\right)^{5/2}$

(C) $w = \left(\frac{a}{P}\right)^{2/5}$

(D) $w = \left(\frac{P}{a}\right)^{2/5}$

7. HARD · DESMOS

For all real numbers x,

$$8^x \cdot 4^{x/3} \cdot \left(\sqrt{2}\right)^x = b^x,$$

where b is a positive constant. Which of the following is equal to 2?

(A) $b^{6/25}$

(B) $b^{25/6}$

(C) b^{25}

(D) $b^{-6/25}$

8. Hard · Desmos

$$\frac{2}{a} - \frac{3}{b} = \frac{5}{c}$$

In the equation above, a, b and c are nonzero constants and $2c \neq 5a$. Which of the following expresses b in terms of a and c?

(A) $b = \dfrac{3ac}{5a - 2c}$

(B) $b = \dfrac{2c - 5a}{3ac}$

(C) $b = \dfrac{3ac}{2c + 5a}$

(D) $b = \dfrac{3ac}{2c - 5a}$

9. Hard · Desmos

The polynomial $6x^4 + kx^2 + 24$, where k is a constant, can be written as $(2x^2 + a)(3x^2 + b)$, where a and b are integers. What is the greatest possible value of k?

10. Hard · Desmos

For a real number a with $|a| \neq 1$,

$$\frac{1}{1-a} + \frac{1}{1+a} + \frac{2}{1+a^2} + \frac{4}{1+a^4} = 10.$$

What is the value of a^8?

(A) $\dfrac{5}{4}$

(B) $\dfrac{9}{5}$

(C) $\dfrac{1}{5}$

(D) $\dfrac{4}{5}$

11. HARD · NO CALCULATOR NEEDED

For real numbers a and b, $a^2 - b^2 = 36$. What is the value of $(3a + b)^2 - (a + 3b)^2$?

12. HARD · DESMOS

If $2x + 3y = 8$, which of the following is equivalent to $2x^2 + 3xy - 4x - 6y$?

(A) $8x - 8$

(B) $16 - 8x$

(C) $8x - 4y$

(D) $8x - 16$

13. HARD · DESMOS

Let t be a positive real number with $t \neq 1$. Define $m = \sqrt[3]{t^2}$ and $n = \sqrt{t}$. If

$$\frac{m^p \cdot n^3}{\sqrt[6]{t}} = t^5,$$

what is the value of p?

14. HARD · DESMOS

The expression $\dfrac{(2x-3)^4(x+2)-7}{5}$ is equivalent to $ax^5+bx^4+cx^3+dx^2+ex+f$, where a, b, c, d, e and f are constants. What is the value of $a+b+c+d+e+f$?

(A) $-\dfrac{4}{5}$

(B) $\dfrac{3}{5}$

(C) -4

(D) 31

15. HARD · DESMOS

The polynomial $12x^4-84x^2+144$ can be written in the form $(ax^2+b)(cx^2+d)$, where a, b, c and d are integers with $a>0$ and $c>0$. What is the least possible value of $a+b+c+d$?

(A) -27

(B) -22

(C) -18

(D) -38

16. HARD · DESMOS

The function f is defined by $f(x)=\dfrac{6^{x+1}\cdot 4^x}{3^{x-2}}$. It can be written in the form $f(x)=a\cdot b^{\,x-1}$, where a and b are positive constants. What is the value of $a+b$?

(A) 62

(B) 440

(C) 456

(D) 3456

17. HARD · DESMOS

For a positive real number m, let $a = \sqrt[3]{m}$ and $b = \sqrt[4]{m}$. If $ab = 128$, what is the value of ab^2?

18. HARD · DESMOS

For all real numbers x, the function g is defined by $g(x) = \dfrac{9^{x+2} \cdot 10^{x+1}}{6^{x-1}}$. The function can be written in the form $g(x) = k \cdot 15^{x+1}$, where k is a constant. What is the value of k?

19. HARD · DESMOS

Positive real numbers a and b, both greater than 1, satisfy $a^3 = b^2$. If $\dfrac{a^9}{\sqrt[3]{b}} = b^k$, what is the value of k?

(A) $\dfrac{17}{3}$

(B) 6

(C) $\dfrac{19}{3}$

(D) $\dfrac{17}{2}$

20. HARD · DESMOS

Positive real numbers x and y satisfy $x \neq y$, $x + y = 15$ and $xy = 36$. What is the value of

$$\frac{\frac{1}{x} - \frac{1}{y}}{\frac{1}{x^2} - \frac{1}{y^2}} \; ?$$

(A) $\frac{5}{12}$

(B) 1

(C) 4

(D) $\frac{12}{5}$

21. HARD · DESMOS

What is the solution set of the equation

$$\sqrt{x + 7} = x - 5?$$

(A) $x = 2$ only

(B) $x = 9$ only

(C) $x = 2$ and $x = 9$

(D) There is no solution.

22. HARD · DESMOS

a and b are positive consecutive integers, and $b > a$. If b is thirteen more than $\sqrt{a}$, what is the value of a?

23. HARD · DESMOS

Let m be an even integer. How many possible values of m satisfy $\sqrt{m+11} \leq 4$?

24. HARD · DESMOS

Real numbers x and y satisfy

$$\sqrt[3]{2^x} \cdot 4^y = 256 \qquad \text{and} \qquad \frac{16^x}{\sqrt[3]{4^y}} = 2^{22}.$$

What is the value of xy?

25. HARD · DESMOS

Positive real numbers x and y satisfy

$$\sqrt{x} - \sqrt{y} = 2 \qquad \text{and} \qquad x + y = 26.$$

What is the value of xy?

26. HARD · DESMOS

The function $m(z) = -\sqrt{z^2 + pz + q}$ has graph passing through $(6, 0)$ and $\left(0, -\sqrt{252}\right)$, where p and q are constants. What is the greatest value of t for which $m(t) = 0$?

27. HARD · DESMOS

The function $h(x) = -\sqrt{x^2 + bx + c}$ has a graph passing through $(2, 0)$ and $\left(0, -\sqrt{314}\right)$, where b and c are constants. What is the distance between the two x-intercepts of the graph?

28. HARD · DESMOS

The function $f(x) = 2\left(3x + 1 - \sqrt{5}\right)\left(3x + 1 + \sqrt{5}\right)$ can be written as $f(x) = ax^2 + bx + c$, where a, b and c are constants. What is the value of $\dfrac{b + c}{a}$?

29. Hard · Desmos

The polynomial function p is defined by $p(x) = -(x+3)^2(x-1)^3(x-4)^2$. Consider the following statements.
I. $p(x) \geq 0$ for all $x < 1$.
II. The graph of $y = p(x)$ crosses the x-axis at $x = 4$.
III. $p(x) \to \infty$ as $x \to -\infty$.
Which of the statements must be true?

(A) I only

(B) I and II only

(C) I and III only

(D) II and III only

30. Hard · Desmos

The function h is defined by $h(x) = \sqrt{x-2} + 1$. The function g is defined by $g(x) = h(x+5)$. What is the domain of g?

(A) $x \geq -3$

(B) $x \geq 2$

(C) $x \geq 7$

(D) $x \geq -5$

31. Hard · No calculator needed

A sequence is defined recursively by $a_1 = 5$ and $a_n = 2a_{n-1} - 3$ for $n \geq 2$. What is the value of a_4?

32. Hard · Desmos

A water-treatment filter reduces the concentration of a contaminant by 12% every hour it operates. If the initial concentration is 250 parts per million (ppm), which of the following functions models the concentration $C(t)$, in ppm, after t hours of operation?

(A) $C(t) = 250(1.12)^t$

(B) $C(t) = 250 - 12t$

(C) $C(t) = 250(0.12)^t$

(D) $C(t) = 250(0.88)^t$

33. Hard · Desmos

In a sequence, the first term is -4. Each term after the first is two less than triple the preceding term. Which term is the first one whose absolute value is greater than 150?

(A) 3

(B) 4

(C) 5

(D) 6

34. Hard · Desmos

The exponential function $f(x) = ab^x$, where a and b are positive constants, satisfies $f(2) = 18$ and $f(5) = 486$. What is the value of $f(4)$?

35. HARD · DESMOS

The size of a bacterial colony, in thousands of cells, is modeled by $V(t) = 500 \cdot 2^{t/2} \cdot 4^{t/3} \cdot 16^{-t/12}$, where t is the number of hours since the start of observation. The size of the colony is multiplied by a constant factor c every 6 hours. What is the value of c?

PART B · SELF-ASSESSMENT — 19 PROBLEMS

36. MEDIUM · NO CALCULATOR NEEDED

What is the solution to $\sqrt{2x+7} = 5$?

37. MEDIUM · DESMOS

The estimated population of a new coastal town t years after 2024 is modeled by

$$P(t) = 5000\,(1.15)^{t}\,.$$

According to the model, by approximately what factor does the population increase every 5 years?

(A) 1.15

(B) 1.75

(C) 2.01

(D) 2.99

38. Medium · Desmos

A car is purchased for \$120,000. Its value depreciates according to

$$V(t) = 120000\,(0.88)^{t},$$

where t is the number of years since purchase. To the nearest thousand dollars, what is the value of the car after 4 years?

(A) \$61,000

(B) \$72,000

(C) \$84,000

(D) \$94,000

39. Medium · No calculator needed

The storage capacity of a battery facility is modeled by

$$E(t) = 1450\,(1.08)^{t},$$

where $E(t)$ is the capacity, in megawatt-hours, and t is the number of years after the facility opened, for $0 \le t \le 15$. Which is the best interpretation of 1450 in this context?

(A) The predicted capacity, in megawatt-hours, 15 years after the facility opened

(B) The estimated capacity, in megawatt-hours, when the facility opened

(C) The annual percent increase in capacity

(D) The number of years it takes the capacity to double

40. Medium · Desmos

A population of bacteria in a lab culture is modeled by $P(t) = 300(1.5)^{t}$, where t is the time in hours. What is the population after 3 hours?

(A) 675

(B) 900

(C) 1,012.5

(D) 1,350

41. Medium · Desmos

The number of subscribers to an online tutoring platform grows according to $N(t) = 500(1.2)^t$, where t is measured in months since launch. Which of the following is closest to the number of subscribers after 5 months?

(A) 1,000

(B) 1,100

(C) 1,244

(D) 1,500

42. Medium · Desmos

The amount of a medication remaining in a patient's bloodstream is modeled by $M(t) = 200(0.85)^t$, where t is the number of hours since the dose was taken and $M(t)$ is measured in milligrams. Which of the following is closest to the amount remaining after 6 hours?

(A) 65.5

(B) 75.5

(C) 85.5

(D) 95.5

43. Medium · Desmos

The population of a small town has grown at an average rate of 2.4% per year since 2010. There were approximately 60,000 residents in 2010. Which of the following functions best models the town's population P, in thousands of residents, t years after 2010?

(A) $P(t) = 60(1.024)^t$

(B) $P(t) = 60(2.4)^t$

(C) $P(t) = 1.024t + 60$

(D) $P(t) = 2.4t + 60$

44. Medium · No calculator needed

Each term in the sequence below is 6 times the previous term. What is the 9th term in the sequence?

$$3,\ 18,\ 108,\ 648,\ \ldots$$

(A) 648×9

(B) 3^9

(C) 3×6^8

(D) 6×3^8

45. Medium · Desmos

The first term of a geometric sequence is 2 and the fourth term is 54. What is the sum of the first five terms of the sequence?

46. Hard · Desmos

The function $P(t) = \dfrac{120}{1 + 5(0.8)^t}$ models the number of active users of an app, in thousands, t weeks after launch, for $0 \le t \le 10$. Consider the following statements.

I. P is increasing on the interval $0 \le t \le 10$.

II. There is a time t in the interval at which $P(t) = 60$.

III. There is a time t in the interval at which $P(t) = 80$.

Which of the statements must be true?

(A) I only

(B) I and II only

(C) II and III only

(D) I, II, and III

47. HARD · DESMOS

The value V, in dollars, of a piece of lab equipment t years after it was purchased is modeled by

$$V(t) = 2400\,(0.64)^{t/2.5}.$$

Which of the following is the best interpretation of the model?

(A) It falls 36% every 2.5 years.

(B) It falls 36% every 5 years.

(C) It falls 64% every 2.5 years.

(D) It falls 2.5% every year.

48. HARD · DESMOS

The number of bacteria in a colony t years after January 1, 2020 is modeled by $A(t) = 300 \cdot 4^{t/3}$. Which of the following gives the number of bacteria m months after January 1, 2023?

(A) $300 \cdot 2^{m/18}$

(B) $1200 \cdot 2^{m/36}$

(C) $900 \cdot 2^{m/18}$

(D) $1200 \cdot 2^{m/18}$

49. Hard · Desmos

$$T(t) = 24 + 66(0.85)^t$$

The function T models the temperature, in degrees Celsius, of a cup of tea t minutes after it is placed in a room. Which of the following is the best interpretation of the number 0.85 in this context?

(A) T falls 15% per minute

(B) the gap $T - 24$ falls 15% per minute

(C) $T(1) = 0.85\,T(0)$

(D) T levels off at 85% of $T(0)$

50. Hard · Desmos

The number of visitors to a national park increased by 25% from 2022 to 2023. From 2023 to 2024, the number of visitors decreased by 12%. If the number of visitors in 2024 was 9,020,000, what was the number of visitors in 2022?

51. Medium · Desmos

A jacket is discounted by 30%. A 5% sales tax is then applied to the discounted price. If the final price paid is \$44.10, what was the original price of the jacket, before the discount?

(A) \$42.00

(B) \$58.80

(C) \$60.00

(D) \$63.00

52. Medium · Desmos

A laptop's price increased by 20% during a promotional period, then decreased by 15% after the promotion ended. If the final price is \$1,020, what was the price at the end of the promotional period, before the decrease?

(A) \$950

(B) \$1,000

(C) \$1,050

(D) \$1,200

53. Hard · Desmos

The number of families visiting a city festival's family zone increased by 40% from Week 1 to Week 2, then decreased by 25% from Week 2 to Week 3. If Week 3 attendance was 8,400 families, how many more families attended in Week 2 than in Week 3?

54. Medium · No calculator needed

Which of the following represents the result of decreasing the quantity y by 12%, where $y > 0$?

(A) $0.88y$

(B) $0.12y$

(C) $y - 12$

(D) $y - 0.12$